THE UNIVERSITY OF AZAD JAMMU AND KASHMIR, MUZAFFARABAD

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LAB:	**03**
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COURSE CODE:	**CS-2203**
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LAB 02: CREATION OF NETWORK TOPOLOGY

1. OBJECTIVE

- To understand the basic definition and core components of a computer network.
- To identify different network devices (nodes) and their specific functions.
- To gain hands-on experience using Cisco Packet Tracer simulation software.
- To build a simple Local Area Network (LAN) topology.
- To manually configure static IP addresses on end devices.
- To verify network connectivity and inspect traffic behavior using Command-Line Interface (CLI) commands.

2. INTRODUCTION

2.1 WHAT IS A COMPUTER NETWORK?

A computer network is a digital telecommunications system that allows interconnected nodes to share resources and data. Computing devices communicate with each other using established data link connections over physical cabling (copper, fiber optics) or wireless media.

2.2 NETWORK DEVICES (NODES)

Device	Function
Router	Connects different logical networks together.
Switch	Connects devices within the same local network.
Firewall	Provides network security and controls incoming/outgoing traffic.
End Devices	Client devices (PCs, Laptops) and Servers that end-users interact with directly.
Access Point	Connects wireless devices to a wired network.

3. METHODOLOGY

3.1 SOFTWARE USED

- Cisco Packet Tracer (Latest Version): Network simulation software used for topology design, device configuration, and traffic analysis.

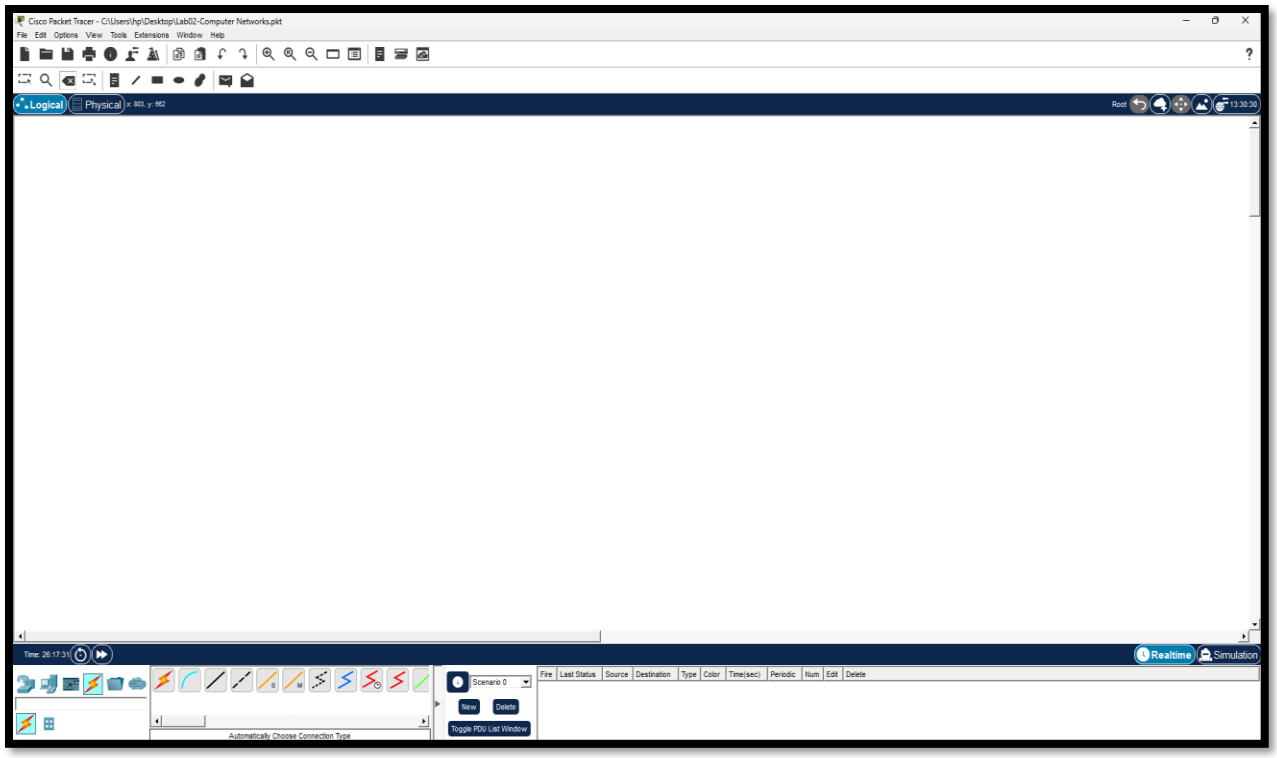
3.2 LAB PROCEDURE

The practical execution was structured across four distinct phases:

1. **Topology Creation:** Assembling physical devices and cable connections.
2. **IP Configuration:** Assigning static logical IP parameters to end devices.
3. **Connectivity Testing:** Executing ICMP ping tests to confirm reachability.
4. **Command Analysis:** Utilizing CLI utilities (ipconfig, tracert, arp, netstat) to analyze network operations

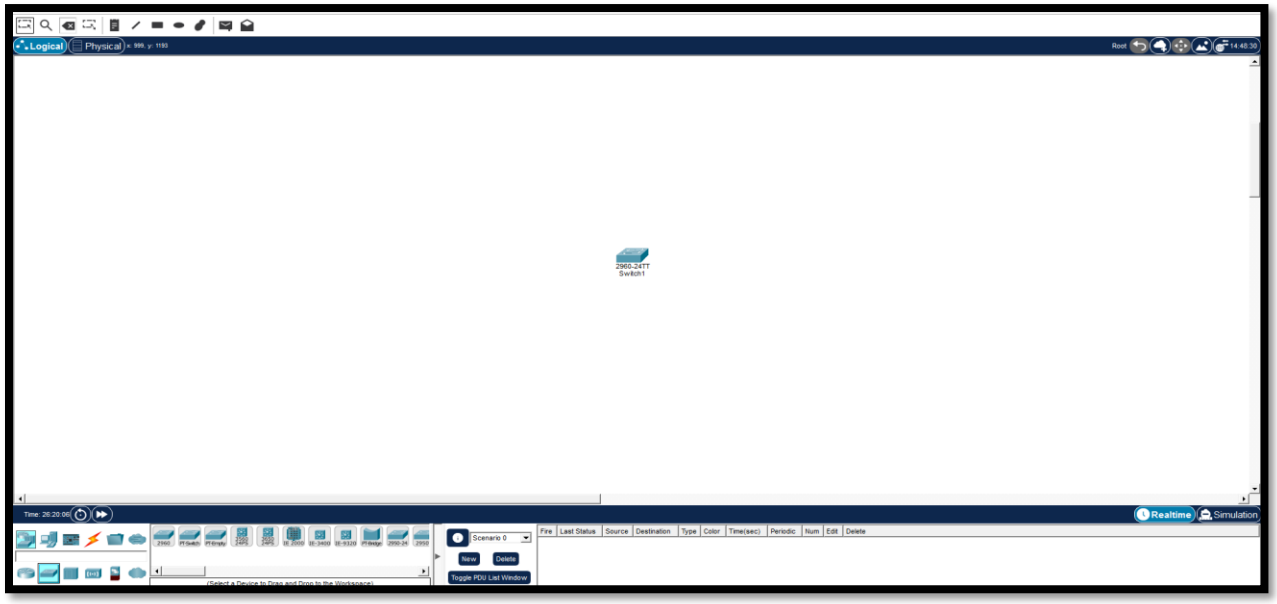
STEP 1: OPENING CISCO PACKET TRACER

Open Cisco Packet Tracer from the system. The main interface appears showing the workspace, device panel, and connection tools.

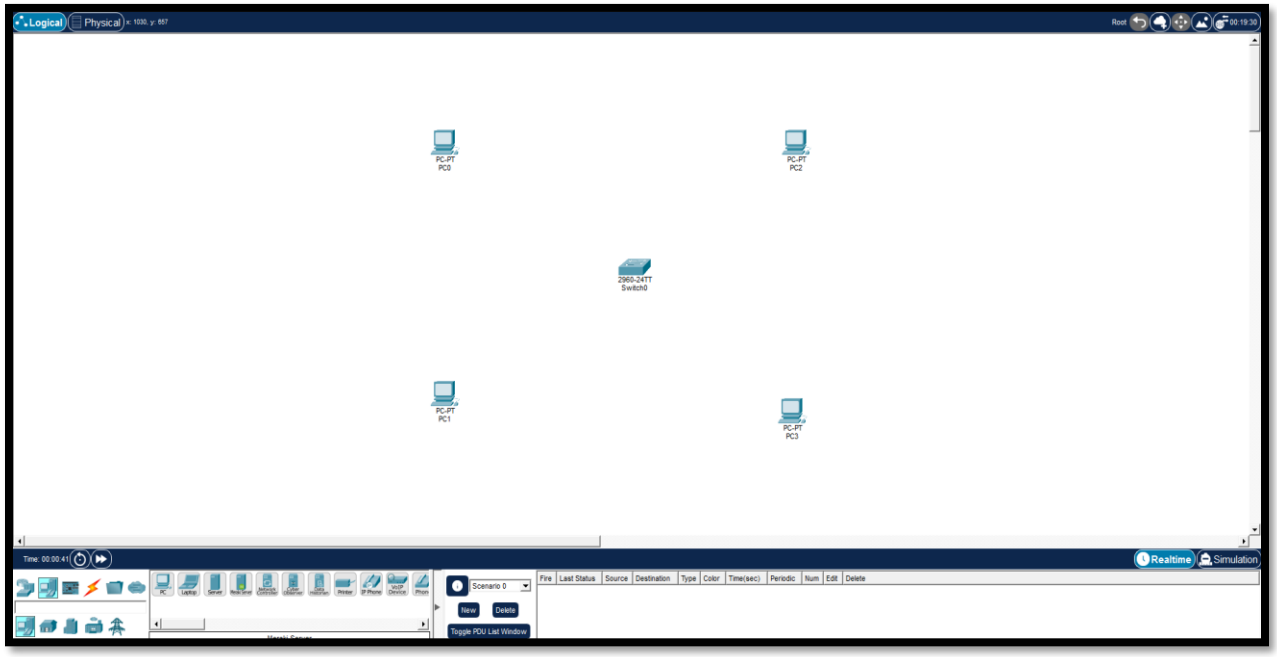


STEP 2: ADDING A SWITCH

From **Network Devices → Switches**, drag a **2960 switch** into the workspace. The switch will be used to connect multiple devices.



STEP 3: ADDING END DEVICES

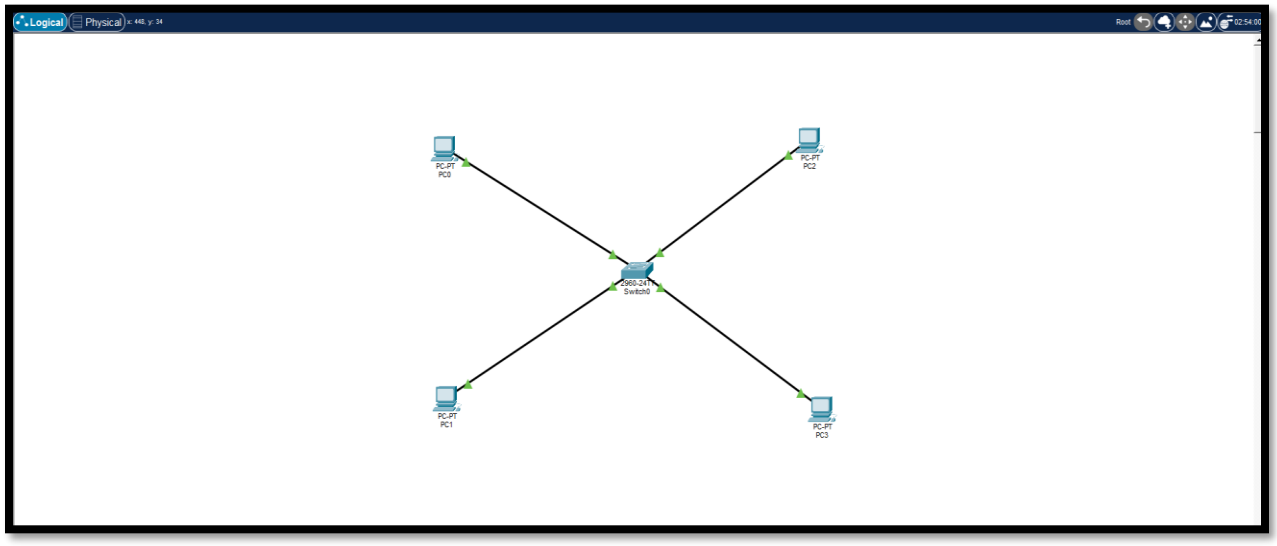


From **End Devices**, drag **two PCs (PC0 and PC1…)** into the workspace. These will act as hosts in the network.

STEP 4: CONNECTING DEVICES

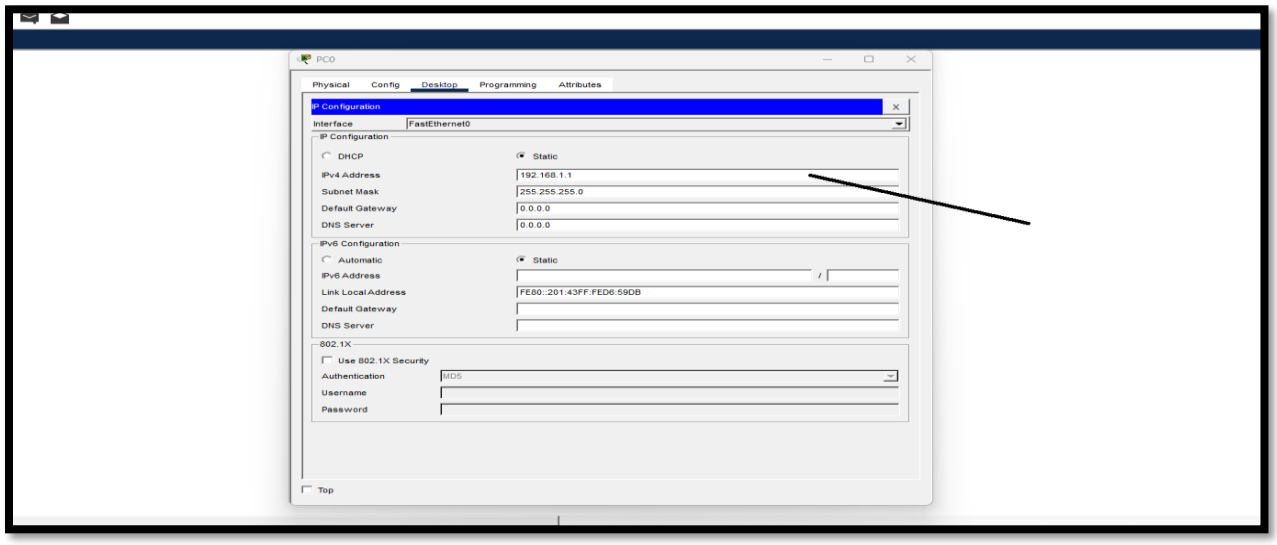
Select **Automatically choose Connection type** from the connections panel.
Connect:

- **PC0 → Switch**
- **PC1 → Switch**



STEP 5: CONFIGURING IP ADDRESS FOR ALL END DEVICES

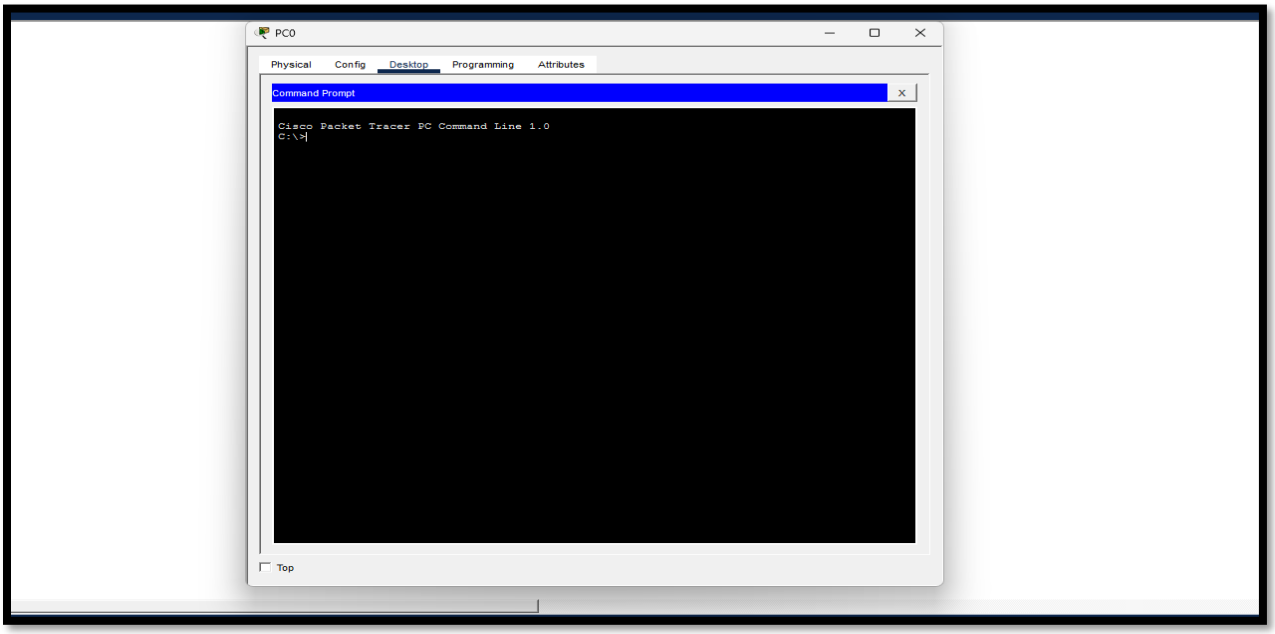
Open PC0 …. P3 → Desktop → IP Configuration.



IP Address: **192.168.1.1 …**
Subnet Mask: **255.255.255.0**

STEP 7: OPENING COMMAND PROMPT

Open **PC0 → Desktop → Command Prompt** to run network commands.



STEP 8: RUNNING IPCONFIG

Run:

ipconfig

This command displays the IP address and subnet mask configuration.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: FE80::201:43FF:FED6:59DB
   IPv6 Address.....: ::
   IPv4 Address.....: 192.168.1.1
   Subnet Mask.....: 255.255.255.0
   Default Gateway.....: ::
                         0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.....: ::
   IPv6 Address.....: ::
   IPv4 Address.....: 0.0.0.0
   Subnet Mask.....: 0.0.0.0
   Default Gateway.....: ::
                         0.0.0.0

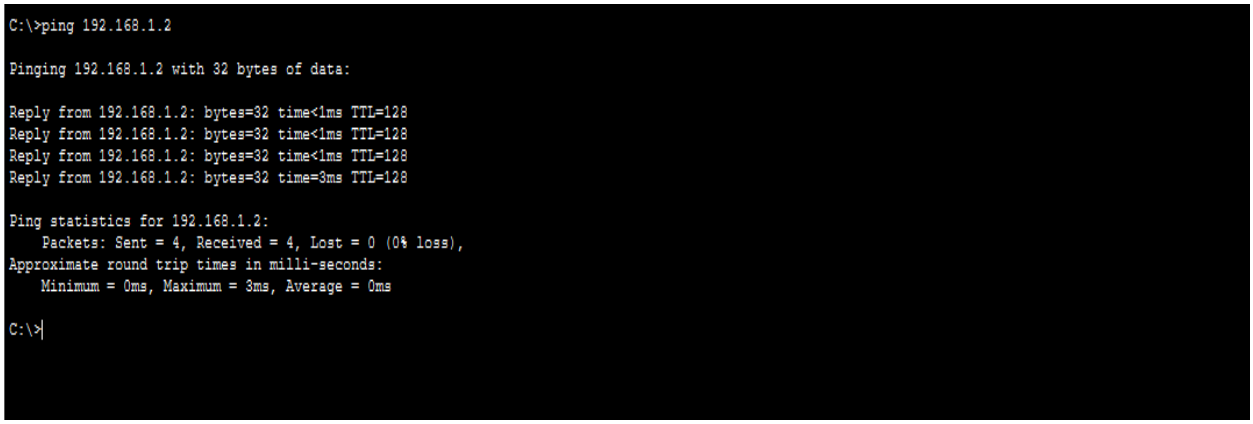
C:\>
```

STEP 9: TESTING CONNECTIVITY USING PING

Run:

ping 192.168.1.2

Successful replies confirm communication between the PCs.



STEP 10: RUNNING TRACERT

Run:

tracert 192.168.1.2

This shows the path packets take to reach the destination.



STEP 11: VIEWING ARP TABLE

Run:

arp -a

This command shows the mapping between IP addresses and MAC addresses.

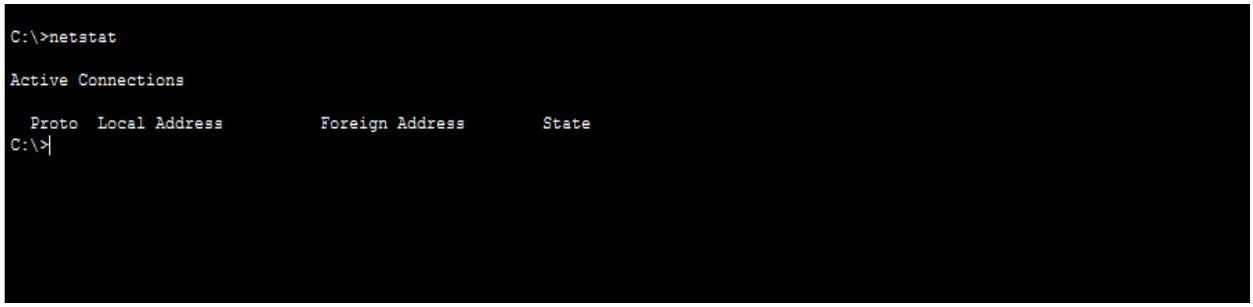


STEP 12: RUNNING NETSTAT

Run:

netstat

This command displays network statistics and active connections.



CONCLUSION

A simple LAN was created using Cisco Packet Tracer by connecting two PCs through a switch. IP addresses were configured, and commands such as ipconfig, ping, tracert, arp, and netstat were used to verify connectivity and analyze the network.